03 · Planning data architecture & MDM

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Planning data architecture and MDM establish an integrated, governed data layer — one coherent model of demand, supply, product, location, and financial data with defined lineage and ownership — so every planning tool reads from a single non-redundant source, instead of a patchwork of point extracts where each tool maintains its own divergent copy of the truth.

2 WHY IT MATTERS

Codd 1970 **(Codd, 1970 — peer-reviewed)**, founding the relational model, established that data should be represented once in an integrated, non-redundant structure rather than duplicated across application-specific files — because redundant copies inevitably diverge and the divergence is where errors and contradictions live. Planning organizations routinely violate this: demand sits in one extract, supply in another, finance in a third, each refreshed on a different cadence and reconciled by hand. Every tool then computes a confident answer from its own slice, and the answers do not agree. The cost is a permanent reconciliation tax and a hard ceiling on maturity — **no amount of advanced analytics or AI rescues a plan built on fragmented, divergent data**, because the sophistication runs on top of a contradiction.

3 HOW IT IS DONE

1. **Define one integrated planning data model.** Establish a coherent model spanning demand, supply, product, location, and financial data, with each entity defined once. The single integrated representation — not a faster set of extracts — is the architectural goal, because non-redundancy is what prevents divergence.
2. **Designate one source of truth per data domain.** For each domain, name the authoritative system and make every tool consume from it rather than from its own copy. Multiple "sources of truth" is a contradiction in terms — and the root of the reconciliation tax.
3. **Establish lineage and ownership.** Track where each data element originates, how it transforms, and who owns it. Lineage makes a divergence traceable to its cause instead of a mystery; ownership ensures someone is accountable for keeping each element correct.
4. **Govern master data as the foundation.** The product, location, and supplier master data that the architecture rests on must be governed with definitions, validation, and a maintenance cadence — the same discipline as operational master-data governance, here applied to the integrated layer the whole platform reads from.
5. **Eliminate the point extracts deliberately.** Each tool-specific extract is a future divergence. Replace them with reads from the integrated layer on a planned schedule, and retire the extracts — tolerating them indefinitely rebuilds the fragmentation the architecture was meant to remove.
6. **Sequence analytics and AI after the data layer.** Advanced analytics, demand sensing, and ML depend on integrated, trustworthy data. Build the data foundation first; analytics deployed on fragmented data produces confident, contradictory output and erodes trust in the very capability the transformation was buying.

> **WORKED EXAMPLE — ILLUSTRATIVE**
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> A company ran demand, supply, and finance from separate extracts reconciled manually — **baseline: three divergent versions of the plan, an analyst team spending much of each cycle reconciling, an AI forecasting pilot that failed because its training data was inconsistent**. It built an integrated planning data model with one source of truth per domain, lineage, and governed master data. Results: **the manual reconciliation collapsed, the plan became one number across functions, and the re-launched analytics pilot finally beat the baseline** — because it ran on integrated data instead of a contradiction.

IN PRACTICE

One model, one source of truth per domain

Design toward a single integrated planning data model with a named authoritative source for each domain, and make tools read from it. This is the architectural commitment that ends the reconciliation tax; everything else in the data track serves this principle.

Lineage and ownership for every element

Document data lineage and assign ownership for each critical element. Lineage turns divergences into traceable defects and ownership ensures they get fixed; together they are what makes the integrated layer trustworthy rather than merely centralized.

Govern the master data the layer rests on

Apply master-data governance — definitions, validation, maintenance cadence — to the product, location, and supplier data the architecture depends on. The integrated layer is only as trustworthy as the master data beneath it; data architecture and MDM are two halves of the same foundation.

Build the data layer before the analytics

Sequence the transformation so the integrated data foundation precedes advanced analytics and AI. Deploying sophisticated capability on fragmented data produces contradictory output and burns the credibility the capability needed; the order — data first, analytics second — is not optional.

WHAT YOU WOULD USE

PLANNING DATA / MDM & INTEGRATION

Cloud data platforms (Snowflake, Databricks, BigQuery) — Modern cloud data platforms host the integrated planning data layer with lineage, governance, and scale. **(Snowflake, Databricks, Google — vendor)** The choice for building one governed source the planning stack and analytics read from.

MDM & integration (Informatica, SAP MDG, Talend) — Master-data management and integration platforms that govern the product, location, and supplier data and pipe it consistently to planning. **(Informatica, SAP, Talend — vendor)** A strong fit where the master-data governance and integration layer is the binding gap.

Data-fabric / lineage & catalog specialists — Dedicated data-catalog, lineage, and data-fabric tools for complex multi-source planning data. **(specialized vendors — vendor)** The vertical option when lineage and governance across many sources is the priority.

For operations that don't need a platform — A single governed planning dataset — even a well-managed central database or warehouse — with one source of truth per domain and documented lineage. The architectural principle, represent each entity once and read from it everywhere, delivers the integrity; the scale of the platform follows the scale of the data, not the principle.

4 SO WHAT

Operations built on one integrated, governed data layer compute a plan that agrees with itself; operations running on point extracts pay a permanent reconciliation tax and hit a maturity ceiling no analytics can break. The fragmented data architecture is the silent ceiling on every planning transformation — the AI pilot fails, the dashboards disagree, the reconciliation never ends, and the cause is always the same contradiction underneath. Integrated data is the unglamorous foundation everything else is built to stand on.

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